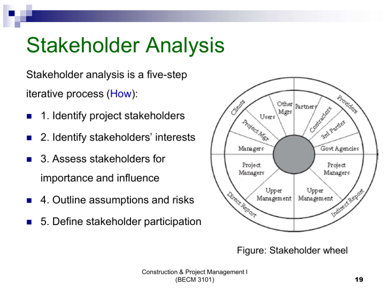
CONSTRUCTION AND PROJECT MANAGEMENT - I

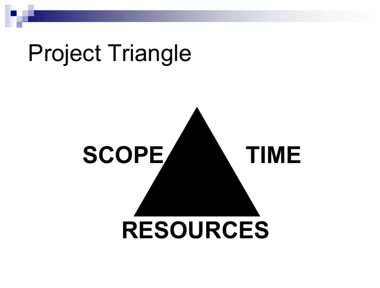
Section B - Lecture 01

INTRODUCTION TO CONSTRUCTION & PROJECT MANAGEMENT

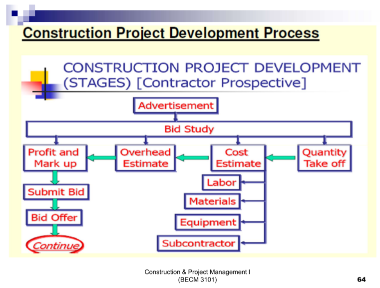
Term-Final Question-to-Slide Mapping | 2016-2024



Stakeholder-analysis process



Three project constraints



Contractor development flow

Scope: The 67-slide lecture was compared against every term-final paper from 2016 through 2024. Original A/B labels are retained in the tables, but classification follows the actual lecture topic because sections changed between years.

Match levels: Exact = almost the whole question is directly answerable; Direct = the main demand is covered but requires synthesis or a small extension; Partial = only one clause or supporting portion is present.

1. Executive Summary

45	26	6	13	482
Question-to-slide links	Exact matches	Direct matches	Partial matches	Total linked marks
All nine exam years	279 linked marks	60 linked marks	143 linked marks	Full marks of linked sub-questions

Most important finding: This is the broadest lecture in the course. It repeatedly generates questions on project definition and constraints, stakeholder analysis, project-manager roles and skills, construction/project management, conceptual design, economic feasibility, development processes and project success.

Year	Linked marks	Relative coverage
2016	46	■■■■■■■■■■■■■■■■■■■■■■■
2017	60	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
2018	67	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
2019	70	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
2020	30	■■■■■■■■■■■■■■■
2021	53	■■■■■■■■■■■■■■■■■■■■■■■■■■■
2022	58	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
2023	48	■■■■■■■■■■■■■■■■■■■■■■■■
2024	50	■■■■■■■■■■■■■■■■■■■■■■■■■

Recurring theme	Main slides	Observed frequency / pattern
Project manager: role, skills, administration, good-PM traits	L40-53	Appears in 2017, twice in 2018, 2019, 2020, 2021, twice in 2022, twice in 2023 and 2024.
Project definition, constraints, SMART and classification	L26-38	Repeated from 2017-2024; triple constraint is a particularly stable question.
Conceptual design and economic feasibility	L55-59	Asked in 2016, 2017, 2019, 2021 and 2023.
Stakeholders and stakeholder analysis	L12-20	Asked in 2019, 2021, 2022 and 2024.
Project development and contractor flow	L54, L60-65	Asked in 2016, 2019 and 2024.
Construction management and objectives	L2-6	Directly asked in 2016, 2017, 2018, 2021 and 2023.

2.1 Detailed Question-to-Slide Mapping: 2016-2018

Year	Original Sec./Q	Marks	Condensed question	Lecture slide(s)	Match	Coverage assessment
2016	A 1(a)	10	Define construction management from different points of view; discuss characteristics of a project plan.	L2-6, L39, L60-63	Partial	Construction management is directly covered, but the lecture does not list a complete set of project-plan characteristics.
2016	A 1(b)	8	Explain the construction project development process with a diagram.	L54, L60-65	Exact	The lecture gives the life-cycle stages, a numbered development process and contractor-perspective flow diagrams.
2016	A 4(a)	10	Functional design process; market-demand and total-cost relationship in economic feasibility.	L55-59	Direct	The economic-feasibility curve is explicit. The design sequence is presented under the term conceptual design process.
2016	A 4(b)	9	Conceptual design process and top-down design style with example.	L57-59	Partial	The six conceptual-design activities are covered; top-down design style is not explained.
2016	A 4(c)	9	Define project planning and write the steps of project planning.	L39, L60-63	Partial	Planning and the development sequence are outlined, but a formal project-planning procedure is not provided.
2017	A 1(a)	8	Define project and project management; write the characteristics of a project.	L26-38	Exact	PMI/Kerzner project definitions, characteristics, constraints and the project-management definition are all present.
2017	A 1(b)	12	How a project manager influences project activities; skills of a good project manager.	L40-53	Exact	The lecture provides responsibilities, management areas, roles, four skill groups and good-PM traits.
2017	A 3(a)	10	Conceptual design process; market demand and total cost as economic feasibility.	L55-59	Exact	Both the design-process diagram and the full economic-feasibility relationship are shown.
2017	A 3(d)	10	Eustress and distress; psychology of a good project manager.	L45-53	Partial	Good-project-manager qualities are detailed, but eustress, distress and the psychology model are absent.
2017	B 5(a)	10	What is construction management? Write its objectives.	L2-6	Exact	Definitions, scope and a complete objective list are provided.
2017	B 5(c)	10	Contribution of construction management to a sustainable building project.	L5-6	Partial	Sustainability appears as an objective, but a detailed contribution framework is not developed.
2018	A 4(b)	10	Contribution of construction management to a sustainable building project.	L5-6	Partial	The lecture establishes sustainability as a CM objective but offers limited supporting detail.
2018	B 5(a)	12	Differentiate construction management and project management; define a project with an example.	L2-6, L26-38	Direct	Separate definitions and scope allow a strong comparison; an explicit comparison table is not supplied.
2018	B 5(b)	8	Explain the triple constraint (time-cost-scope) and its effect on a project.	L29-34	Exact	Scope, time and resources/cost are developed into the project triangle and customer relationship.
2018	B 5(c)	15	Four crucial project-manager skills and how the role influences project activities.	L40-53	Exact	The lecture explicitly groups skills into organization/management, technical, people/communication and administration.
2018	B 6(c)	12	Eustress and distress; psychology of a good project manager.	L45-53	Partial	The good-PM portion is supported; stress concepts are not covered.
2018	B 7(c)	10	How the role of a project manager influences activities; skills of a good project manager.	L40-53	Exact	A direct repeat of the lecture role-and-skill material.

QB = original question-bank PDF page. Slide numbers refer to the uploaded 67-page lecture PDF.

2.2 Detailed Question-to-Slide Mapping: 2019-2021

Year	Original Sec./Q	Marks	Condensed question	Lecture slide(s)	Match	Coverage assessment
2019	A 1(a)	10	What is a project? Outline its features with an example.	L26-35	Exact	Several definitions and six project characteristics directly answer the question.
2019	A 1(b)	7	Who are construction-project stakeholders and why are they important?	L12-20	Exact	Stakeholder definitions, categories, duties and analysis purpose are all included.
2019	A 1(c)	8	Why projects need planning and what planning involves for success.	L39, L60-67	Direct	Planning is defined and linked to development and success factors; the answer needs synthesis across slides.
2019	A 1(d)	10	Explain the construction project development process with a diagram.	L54, L60-65	Exact	Numbered steps and visual flows directly match the requested answer.
2019	A 2(a)	13	Common project stresses and psychology of a good project manager for success.	L45-53, L67	Partial	Successful-PM traits are covered, while the sources and psychology of stress are not.
2019	A 2(c)	10	Triple constraint and its effect on a project, with a diagram.	L29-34	Exact	The project-triangle diagrams and the three constraints directly support the response.
2019	A 3(b)	12	Conceptual design process; market demand and total cost in economic feasibility.	L55-59	Exact	The complete repeated topic is present.
2020	A 1(a)	8	Define project with an example; characteristics of a project and project plan.	L26-39	Direct	Project definition and characteristics are exact; project-plan characteristics require supplementation.
2020	A 1(b)	10	Triple constraint (time-cost-scope) and its effect on a project.	L29-34	Exact	Directly addressed by the constraint and project-triangle slides.
2020	A 1(c)	12	Project-manager influence on activities and skills of a good project manager.	L40-53	Exact	Responsibilities, roles and skill sets are comprehensively covered.
2021	A 1(a)	12	Differentiate construction management and project management; state the SMART rule.	L2-6, L36, L38	Direct	The separate definitions and exact SMART components support the answer; the comparison must be synthesized.
2021	A 1(b)	8	Define a project according to PMI and classify it.	L26, L35, L37	Exact	The PMI definition and project-classification chart are explicitly included.
2021	A 1(c)	15	PESTLE model for identifying stakeholders; how and why stakeholder analysis is performed.	L12-20	Exact	PESTLE, stakeholder categories, five analysis steps and six purposes appear together.
2021	A 3(b)	10	Functional design process; market demand and total cost as economic feasibility.	L55-59	Direct	Economic feasibility is exact; the lecture labels the design sequence conceptual rather than functional.
2021	A 4(a)	8	Common project stresses and psychology of a good project manager.	L45-53	Partial	Only the good-project-manager component is covered.

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2.3 Detailed Question-to-Slide Mapping: 2022-2024

Year	Original Sec./Q	Marks	Condensed question	Lecture slide(s)	Match	Coverage assessment
2022	A 1(a)	14	Common project stresses and psychology of a good project manager for success.	L45-53, L67	Partial	Traits and skills are available; stress theory is outside this lecture.
2022	A 1(b)	13	Stakeholders in a construction project and their roles in the construction sector.	L12-17	Exact	The lecture classifies stakeholders and sets out client, consultant and contractor duties.
2022	A 1(c)	8	Successful project manager toolbox and measurement of project success.	L45-53, L66	Exact	Both the toolbox/skills and six success measures are provided.
2022	A 2(c)	13	Project-manager influence on project activities and skills of a good project manager.	L40-53	Exact	A recurring exact match.
2022	A 4(b)	10	Triple constraint and its effect on a project, with a diagram.	L29-34	Exact	Directly illustrated by the project triangle.
2023	A 1(a)	10	Define construction management and state the main criteria for a proposed industrial building.	L2-8, L22-24	Partial	CM and broad construction-type/financing context are covered, but a complete industrial-building criteria list is not.
2023	A 1(b)	15	Administration skills needed by a good project manager and how the role influences activities.	L40-53	Exact	Administration receives three detailed slides and is integrated with role, management and communication duties.
2023	A 1(c)	10	Conceptual design process; market demand and total cost in economic feasibility.	L55-59	Exact	A repeated exact topic.
2023	A 3(a)	13	Common project stresses and psychology of a good project manager for success.	L45-53, L67	Partial	The good-PM and success portions are covered; stress theory is absent.
2024	A 1(a)	11	Define a project using the three constraints; explain the SMART rule.	L26-36	Exact	Project definitions, scope-time-resources triangle and all SMART components are provided.
2024	A 1(b)	12	Key factors for successful projects; contractor-perspective project-development flow diagram.	L64-67	Exact	The exact contractor flow and seven success factors appear consecutively.
2024	A 1(c)	12	Construction-project stakeholders; how and why stakeholder analysis is performed.	L12-20	Exact	The lecture directly supplies the stakeholder list, method and purposes.
2024	A 3(a)	15	Eustress and distress; psychology of a good project manager for success.	L45-53, L67	Partial	Good-PM qualities and project-success factors are covered; eustress/distress are not.

QB = original question-bank PDF page. Slide numbers refer to the uploaded 67-page lecture PDF.

3. Slide-to-Question Reverse Index

Slides	Core topic	Mapped questions	Priority
L2-6	**Construction management: meaning, scope, activities and objectives**	2016 A1(a); 2017 B5(a), B5(c); 2018 A4(b), B5(a); 2021 A1(a); 2023 A1(a)	High
L12-20	**Stakeholders, PESTLE, stakeholder-analysis method and purpose**	2019 A1(b); 2021 A1(c); 2022 A1(b); 2024 A1(c)	Very high
L26-38	**Project definitions, characteristics, constraints, triangle, SMART, classification and PM definition**	2017 A1(a); 2018 B5(a-b); 2019 A1(a), A2(c); 2020 A1(a-b); 2021 A1(a-b); 2022 A4(b); 2024 A1(a)	Very high
L39-53	**Planning, project-manager responsibilities, roles, skills, toolbox, administration and good-PM traits**	2016 A1(a), A4(c); 2017 A1(b), A3(d); 2018 B5(c), B6(c), B7(c); 2019 A1(c), A2(a); 2020 A1(c); 2021 A4(a); 2022 A1(a,c), A2(c); 2023 A1(b), A3(a); 2024 A3(a)	Highest
L54-59	**Life cycle, economic feasibility, conceptual-design process**	2016 A4(a-b); 2017 A3(a); 2019 A3(b); 2021 A3(b); 2023 A1(c)	Very high
L60-65	**Project-development process and contractor-perspective flow**	2016 A1(b), A4(c); 2019 A1(c-d); 2024 A1(b)	High
L66-67	**Project-success measures and key success factors**	2019 A1(c), A2(a); 2022 A1(a,c); 2024 A1(b), A3(a)	High

Fastest revision route: L40-53 -> L26-38 -> L54-65 -> L12-20 -> L2-6 -> L66-67. This order prioritizes the most repeated and highest-mark material.

4. Exam-Focused Answer Construction Guide

1. Project, constraints and SMART

Start with the PMI/Kerzner definition (L26-28). State temporary, unique, resource-consuming and uncertain characteristics (L35). Draw the scope-time-resources triangle (L32-34). Finish with Specific, Measurable, Acceptable, Realistic and Time-based objectives (L36).

2. Stakeholders and analysis

Define stakeholder (L12), classify contractual/non-contractual groups (L13), explain client-consultant-contractor duties (L14-17), sketch PESTLE (L18), then list the five analysis steps and six purposes (L19-20).

3. Project-manager role and skills

Define accountability and lifecycle responsibilities (L40). Mention managed domains (L41), roles (L42-44), toolbox and four skill families (L45-52), and conclude with good-PM traits (L53).

4. Conceptual design and feasibility

Draw the process diagram (L57). Explain formulation, analysis, search, decision, specification and modification (L58-59). Draw the total-cost curve, identify F and H, and explain increasing/decreasing returns to scale (L55-56).

5. Project development process

Use the lifecycle stages (L54), numbered owner-side sequence (L60-62), then the complete flow/gates (L63). For contractor perspective, reproduce L64-65 and end with cost/time/quality control.

6. Construction management

Separate construction technology from construction management (L2), define CM (L3), compare owner and builder perspectives (L4), list main activities (L5), then state objectives including economy, quality, safety, value, client satisfaction and sustainability (L6).

5. Coverage Limits, Secondary Links and Final Priority

Question area	Lecture support	What is still needed
Eustress, distress, common stresses and Plutchik model	Good-PM skills and traits only (L45-53, L67)	Stress definitions, emotion model and project-stress causes must come from another lecture.
Detailed project-planning method / Engineering V-concept	High-level planning and development sequence (L39, L60-63)	Formal planning steps, planning challenges and V-model are not supplied.
Project-plan characteristics	Project definition, planning and development context	A separate list of plan characteristics is not stated.
Sustainable construction-management contribution	Sustainability appears as a CM objective (L6)	Examples, strategies and detailed contribution mechanisms are needed.
Industrial-building selection criteria	Construction types, nature and financing context (L7-8, L22-24)	A project-specific industrial-site/building criteria framework is absent.
Project organization, WBS/OBS, project team and teamwork	Only indirect references	These should be mapped to later Section B lectures rather than this introduction lecture.

Study order	Slides	Focus
Priority 1	L40-53	Roles, skills, administration and good-PM traits
Priority 2	L26-38	Project definition, characteristics, constraints, SMART and classification
Priority 3	L54-65	Lifecycle, design, feasibility and development processes
Priority 4	L12-20	Stakeholders, PESTLE and stakeholder analysis
Priority 5	L2-6 and L66-67	Construction management plus project-success criteria

Bottom line: Mastering the exact/direct sections of this lecture prepares 32 of the 45 linked question parts. Partial links are useful for structure, but should not be treated as complete answers.